



INTERNATIONAL
HELLENIC
UNIVERSITY

Prof. Periklis Chatzimisios (pchatzimisios@ihu.gr)

IoT Verticals

Smart
Wearables



Smart
Home



Smart
City



Smart
Agriculture



Connected
Car



Health
Care



Industry
Automation

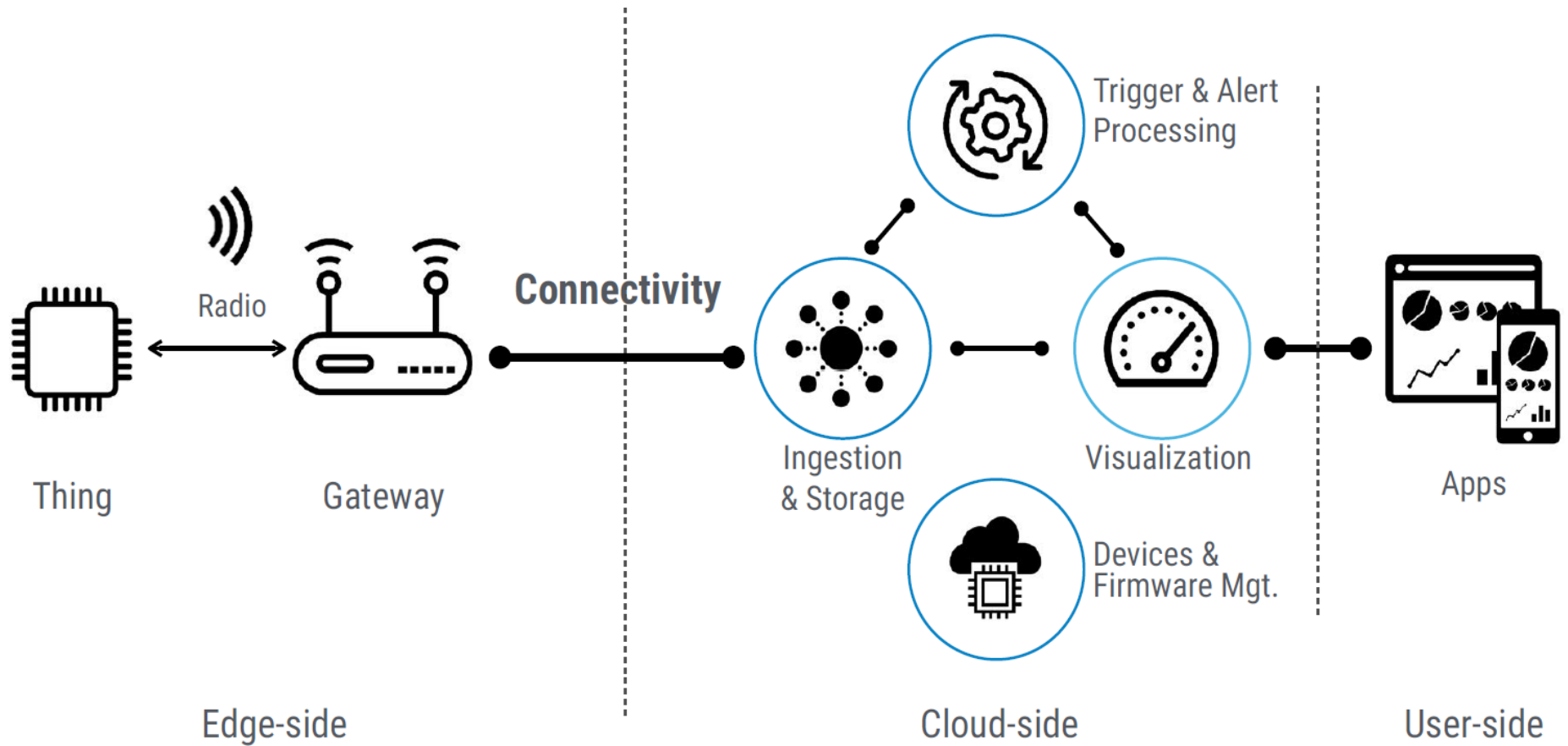


Smart
Energy



Source: Keysight Technologies

IoT common architecture



Three main approaches for networking



Short-range multihop

- ZigBee
- WiFi low energy
- RFID



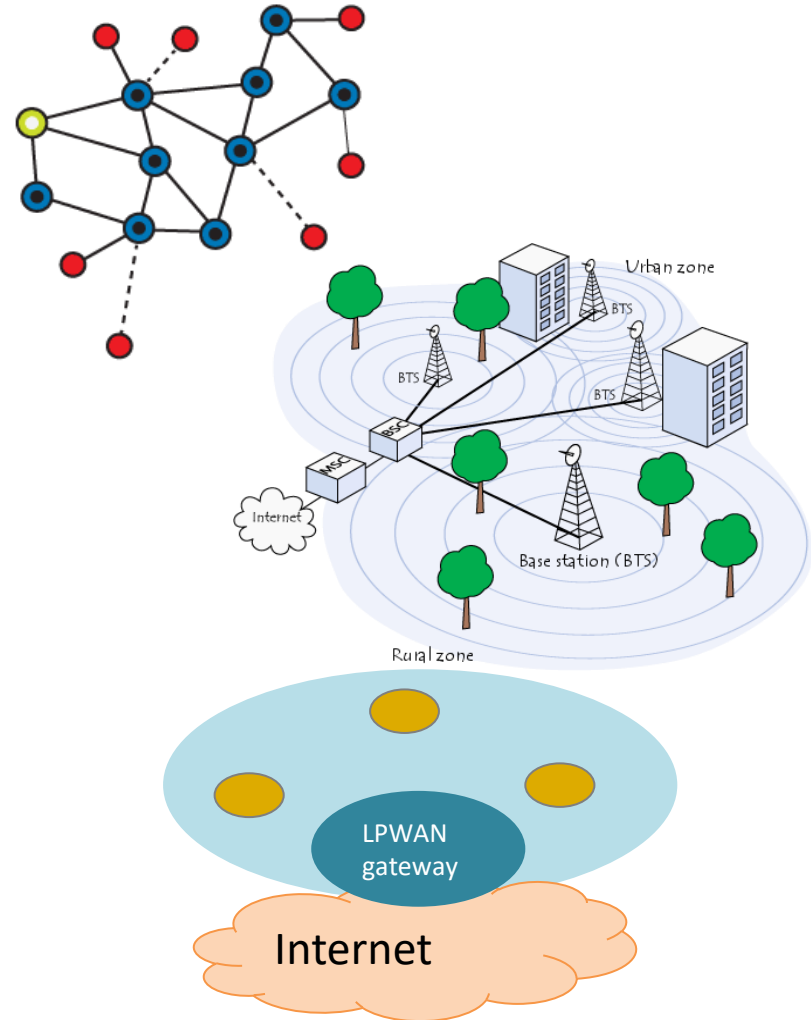
Cellular

- GSM
- LTE-A
- 5G

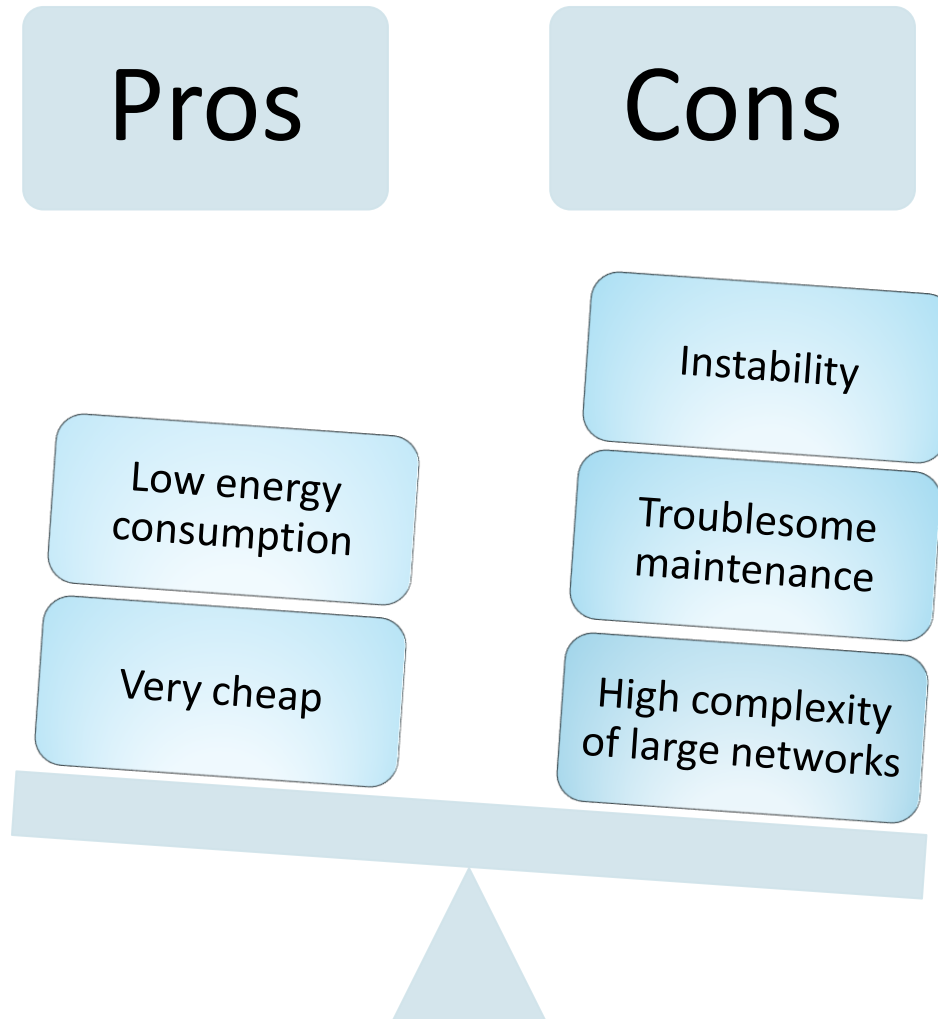


Low Power Wide Area Networks (LPWAN)

- SIGFOX
- NB-IoT
- LoRa



Short range multihop



Cellular-based solutions

Pros

Easy integration with
rest of the world

Well-established
technology

Almost ubiquitous
coverage

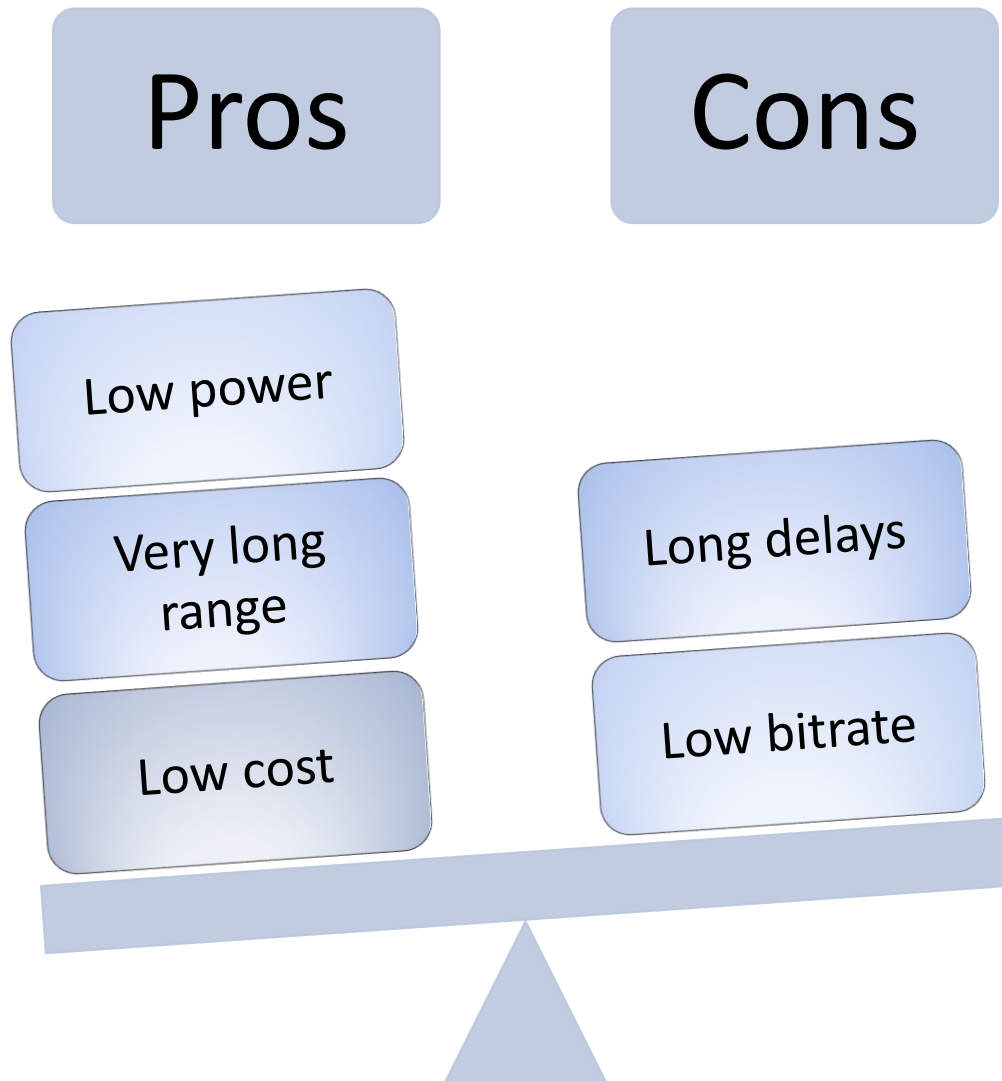
Long range

Cons

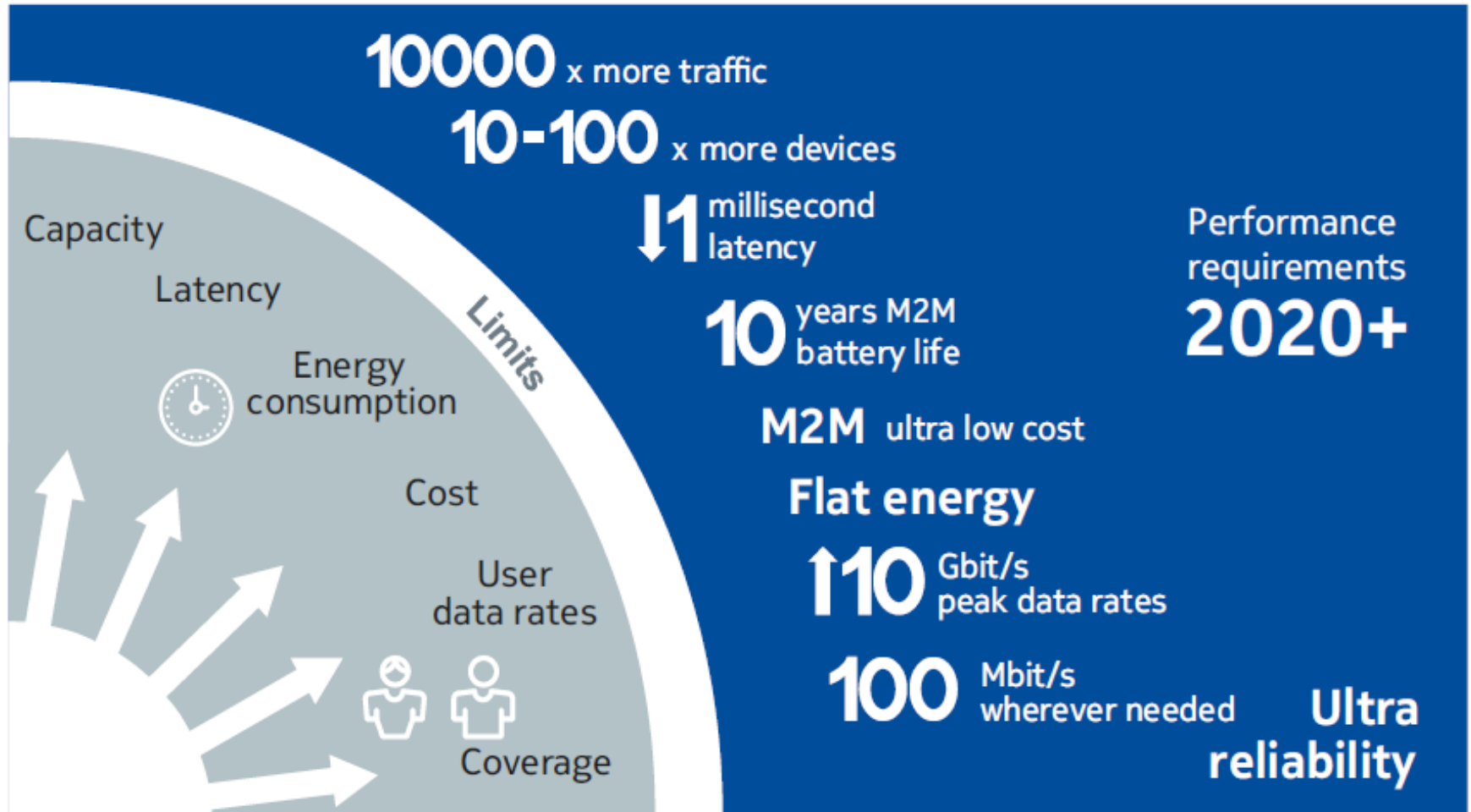
Architectural limits

Energy efficiency

Costs



5G requirements



Source: *Understanding 5G*, Anritsu whitepaper, 2016

Key Performance Indicators – ITU (1)

Peak data rate

Maximum achievable data rate under ideal conditions per user/device (in Gbit/s).

User experienced data rate

Achievable data rate that is available ubiquitously across the coverage area to a mobile user/device (in Mbit/s or Gbit/s).

Latency

The contribution by the radio network to the time from when the source sends a packet to when the destination receives it (in ms).

Mobility

Maximum speed at which a defined QoS and seamless transfer between radio nodes which may belong to different layers and/or radio access technologies (multi-layer/-RAT) can be achieved (in km/h).

Connection density

Total number of connected and/or accessible devices per unit area (per km²).

Energy efficiency

Energy efficiency has two aspects:

- on the network side, energy efficiency refers to the quantity of information bits transmitted to/received from users, per unit of energy consumption of the radio access network (RAN) (in bit/Joule);
- on the device side, energy efficiency refers to quantity of information bits per unit of energy consumption of the communication module (in bit/Joule).

Spectrum efficiency

Average data throughput per unit of spectrum resource and per cell (bit/s/Hz).

Area traffic capacity

Total traffic throughput served per geographic area (in Mbit/s/m²).



7

Billion Devices

2014

Throughput

but there is more

500

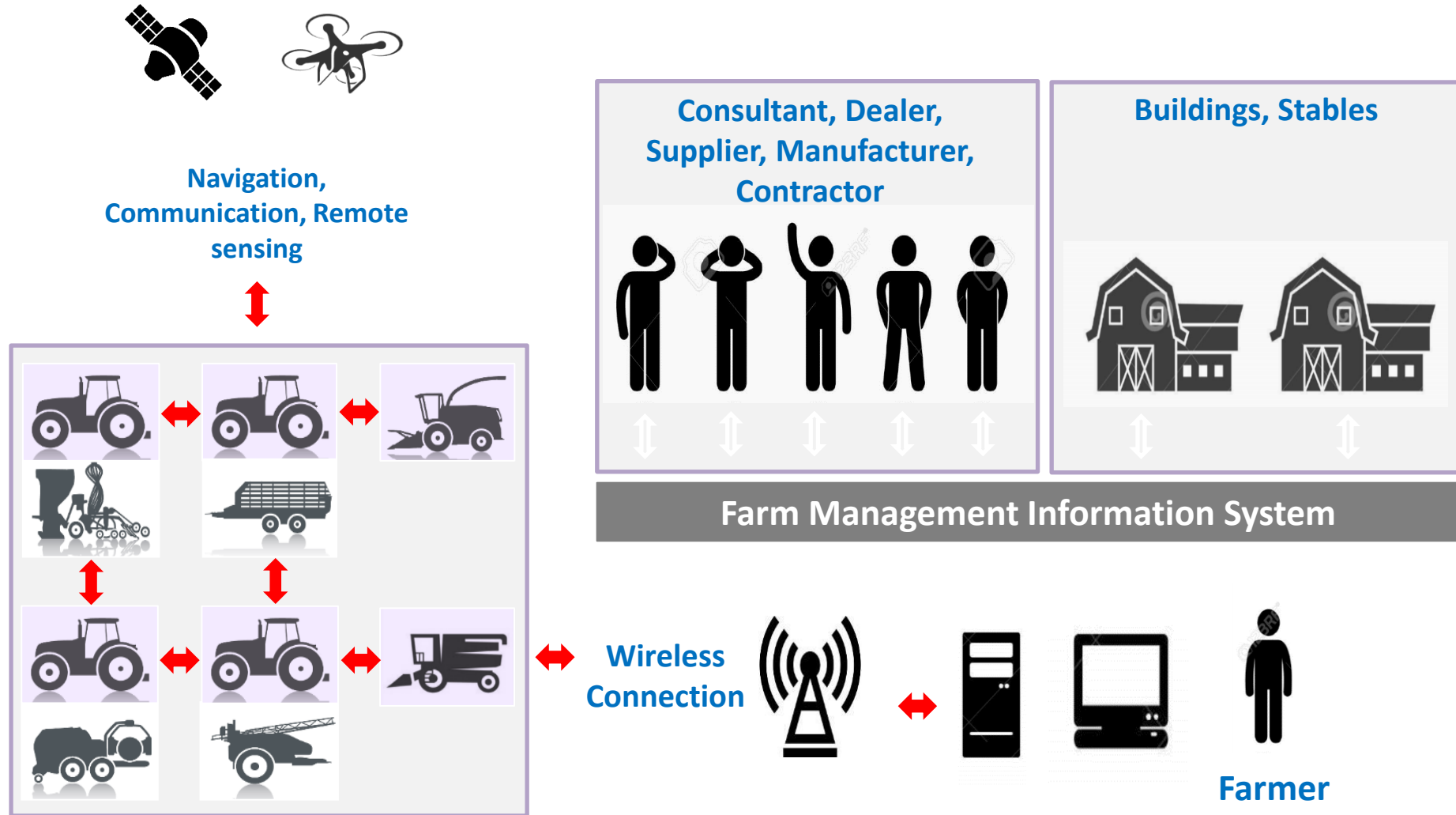
Billion Devices

2022

Reliable Multicast



Precision Farming/Agriculture

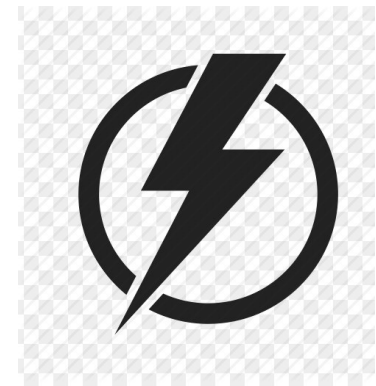


Industrial Internet of Things (IIoT)

Industrial Internet of Things: When the IoT is extended for industrial applications such as manufacturing and supply chain management.

IIoT is an industrial application of IoT (a network connecting industrial machinery and devices) in areas such as oil and gas, transportation, mining and metals, mining, aviation, logistics, etc.

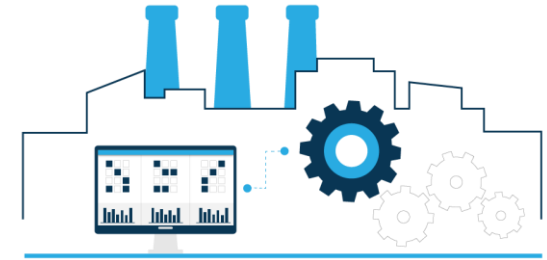
- **Predictive Maintenance**
- **Adaptive Assembly Lines / Customized Products**



Characteristics of IIoT (1)

IIoT : an application of IoT

IIoT key enabler for Industry 4.0



Requirements of Industrial Communications

- **Guarantees → Determinism**
 - **High Reliability**: almost no packet loss
 - **Delay**: deadlines (real time)

Wired Industrial Networks



Wireless Sensor & Actuators Networks

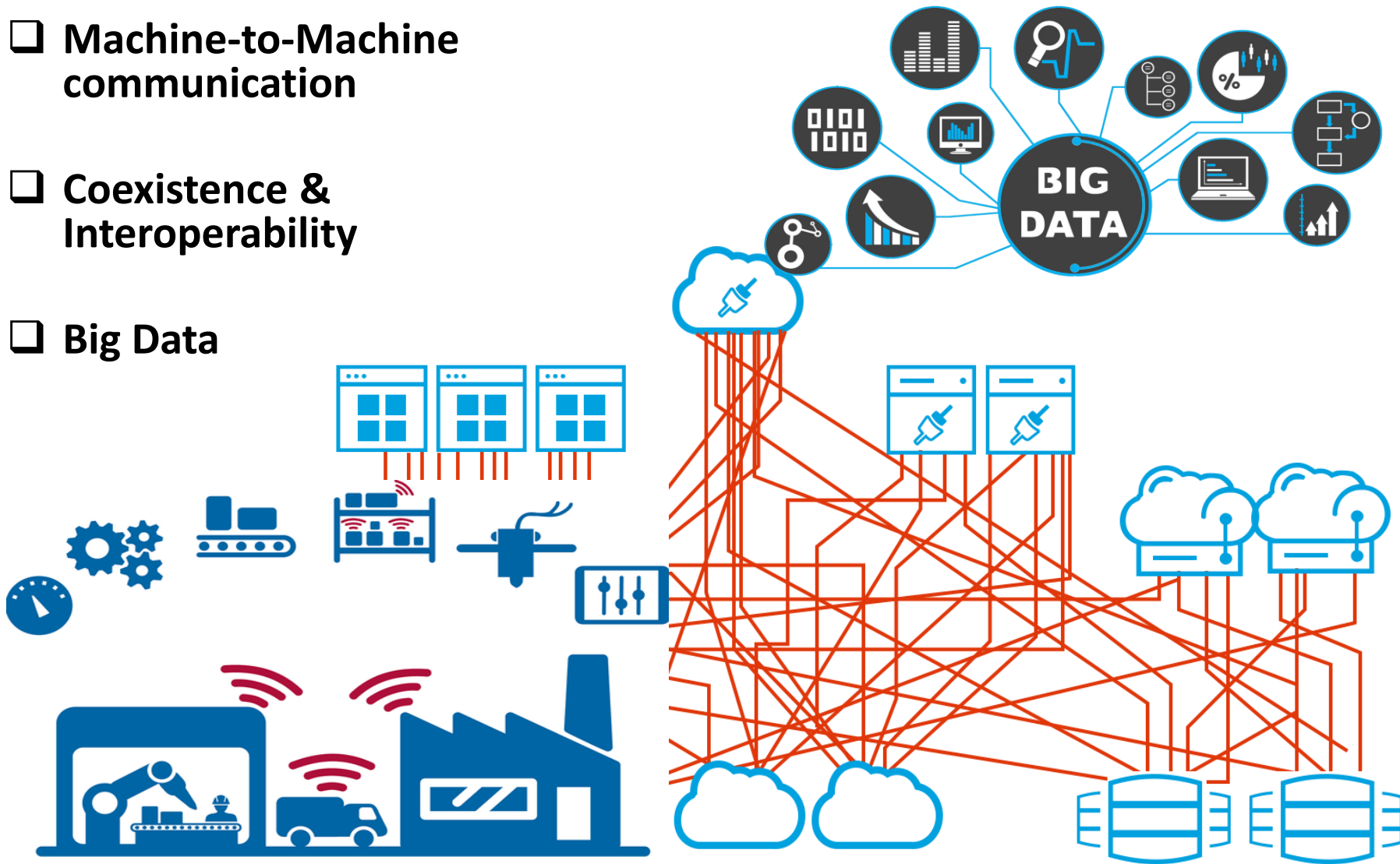
- Reliable
- Low delay
- High data transfer speed
- Not flexible
- Costly

- Unreliable
- No deterministic delay
- Flexible, reconfigurable
- Cost reduction
- Safety

Best effort

Characteristics of IIoT (2)

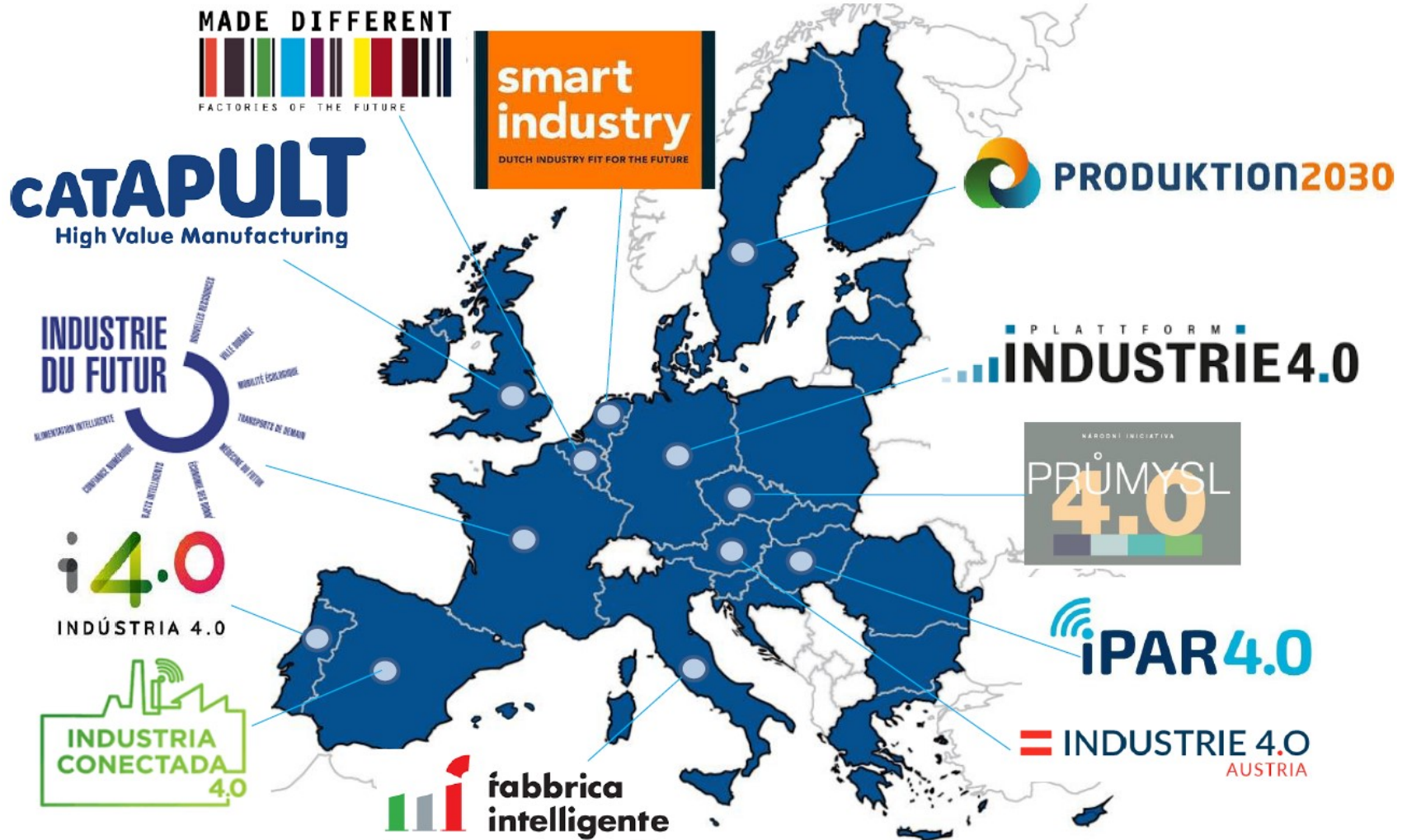
- ❑ Machine-to-Machine communication
- ❑ Coexistence & Interoperability
- ❑ Big Data



Industry 4.0



Industry 4.0 in European Union



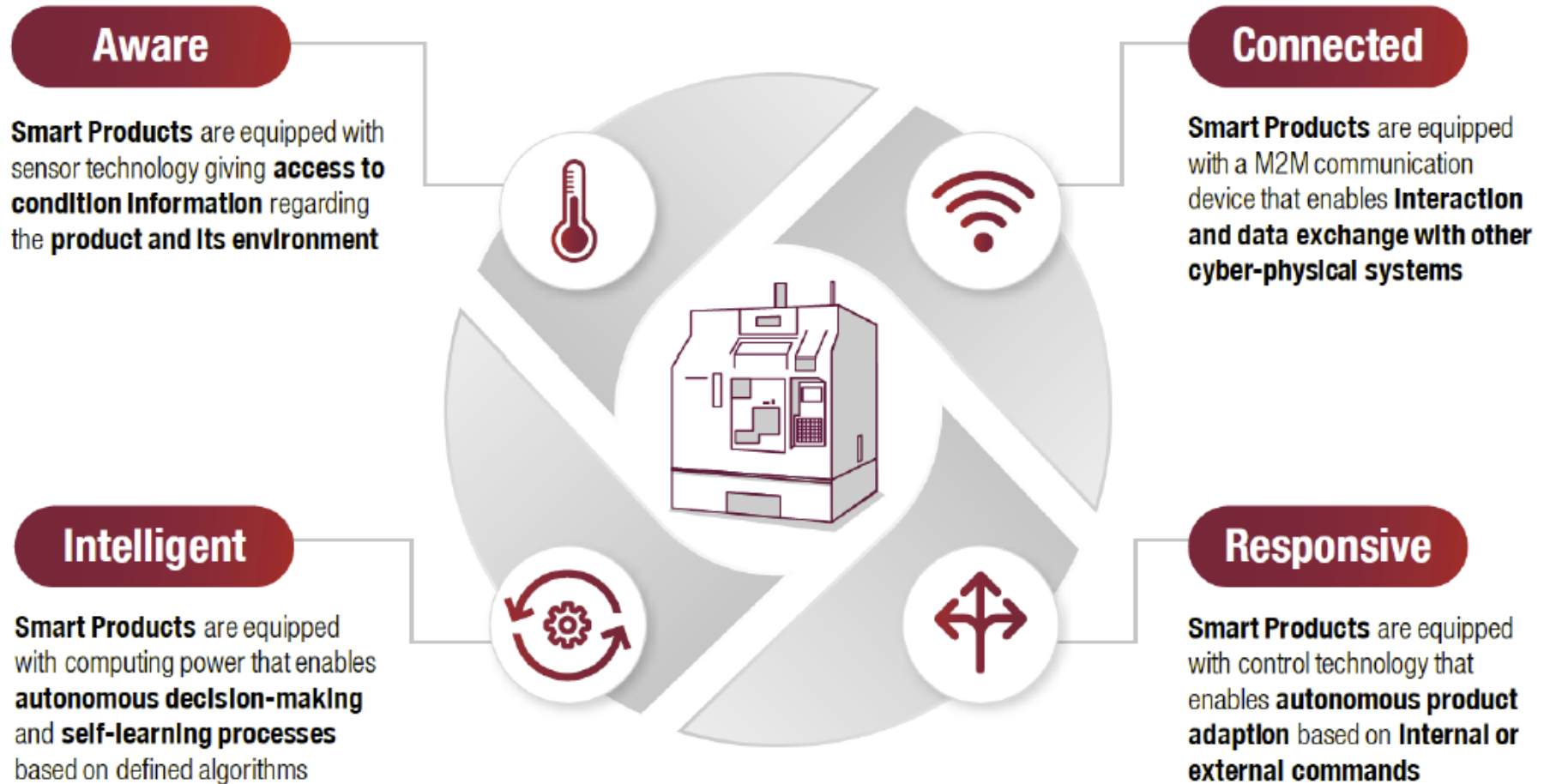
Industry 4.0 in USA and China



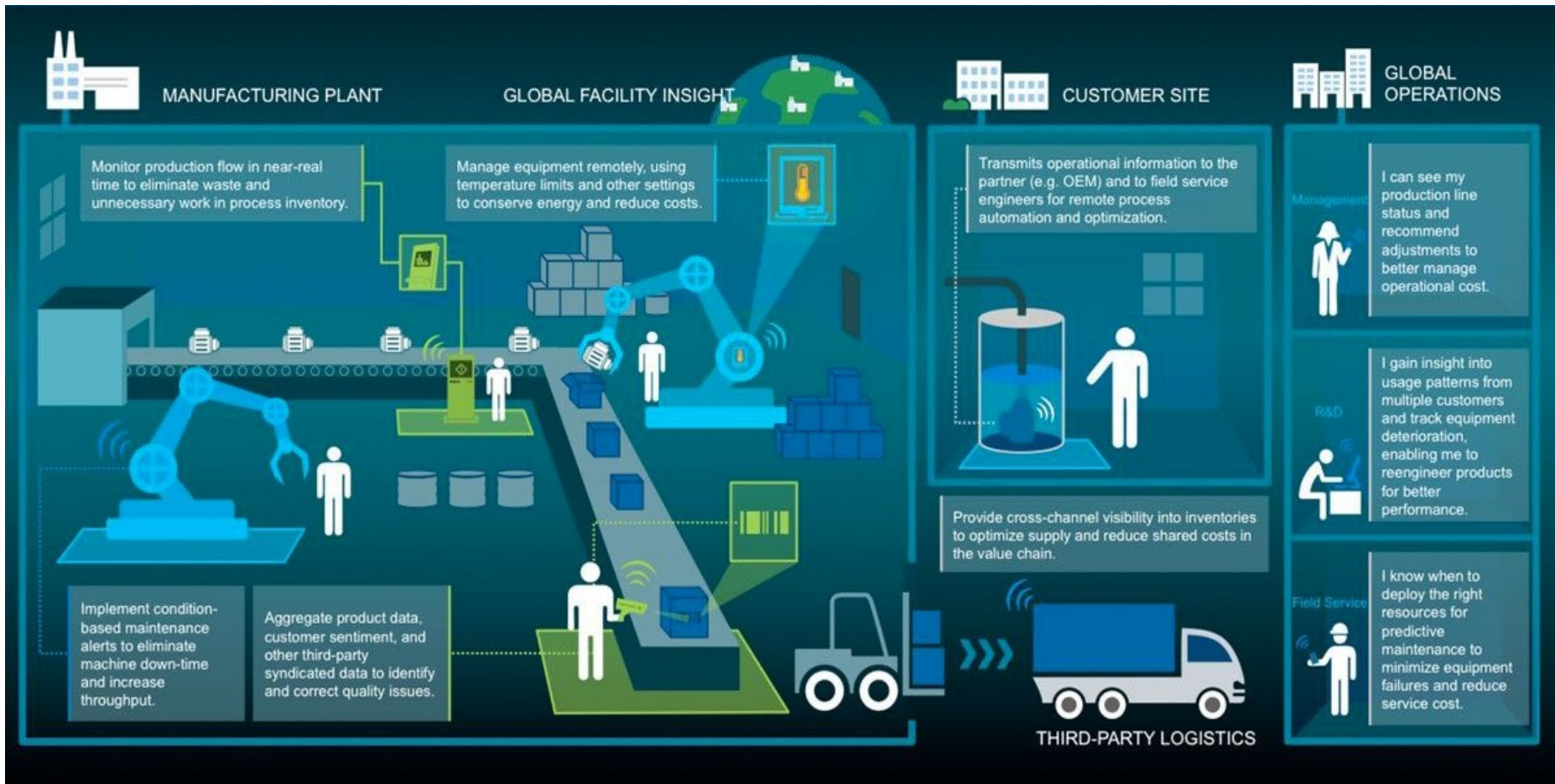
National Network for Manufacturing Innovation (NNMI)

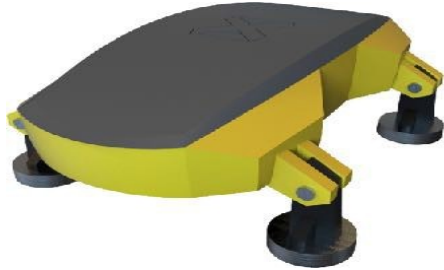


Industry 4.0 - Main characteristics



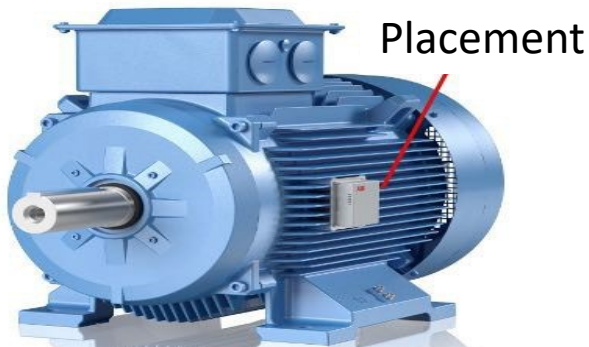
Industrial IoT - Manufacturing



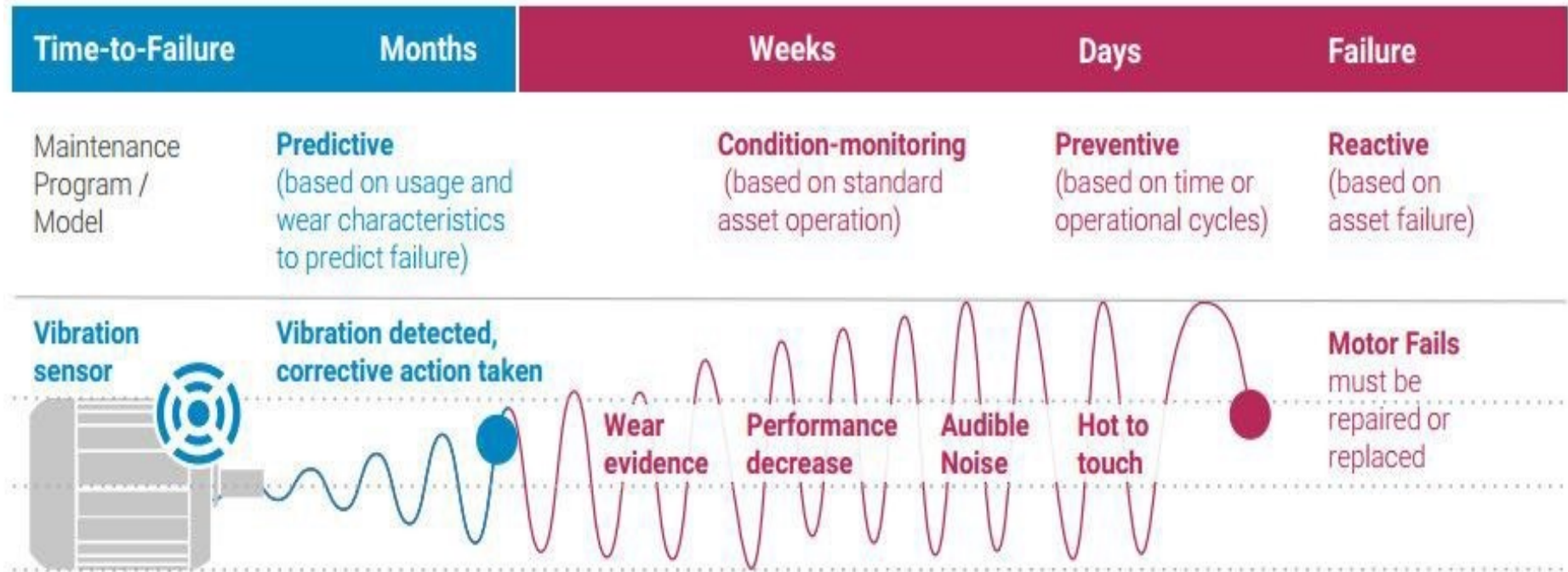


Motor vibration sensor:

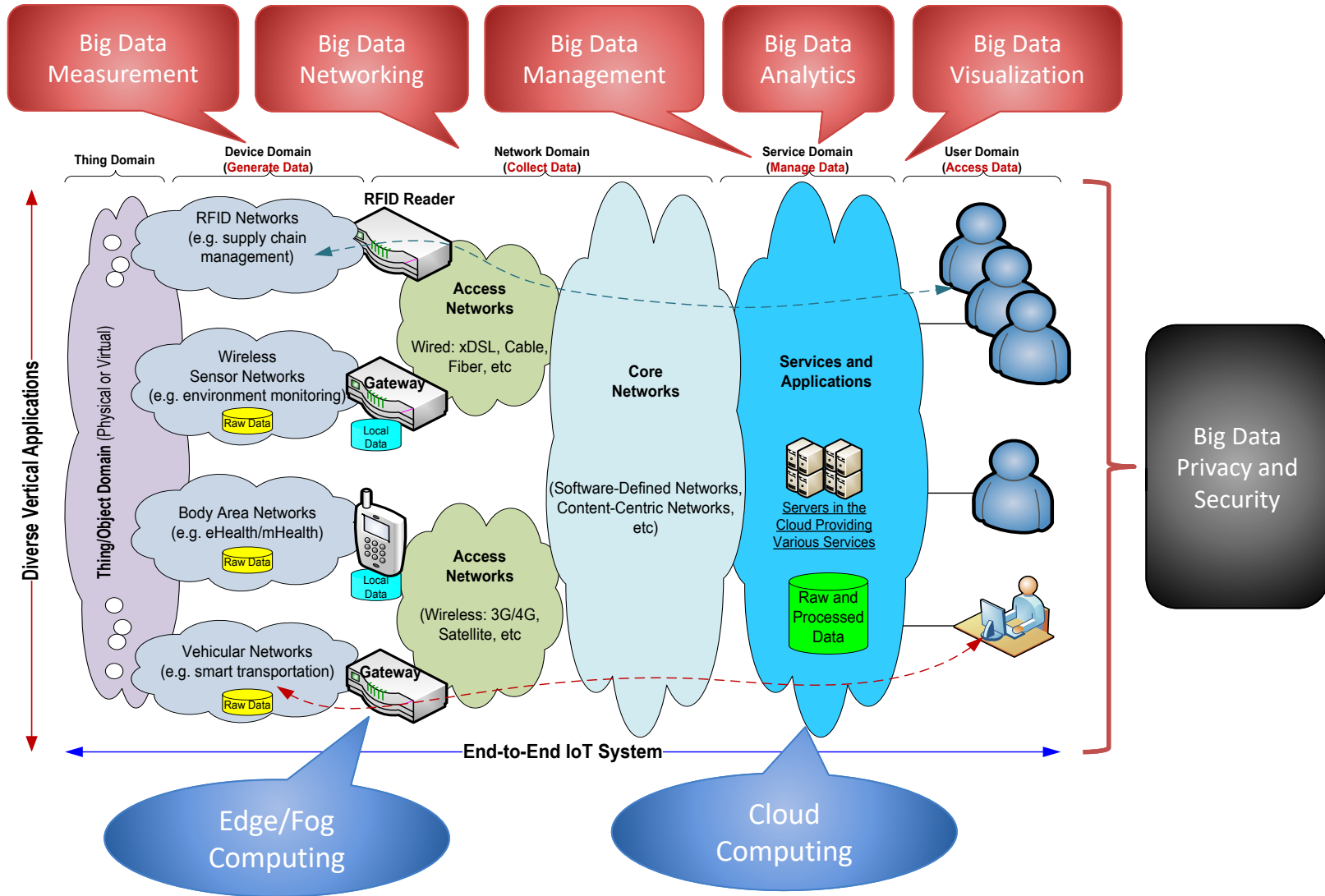
- Built for Industrial IoT, for preventive maintenance
- Measure vibration behavior on machinery's motors
- Detect anomalies and trigger alert
- Suggest maintenance through Machine Learning
- Energy harvesting from vibration and/or heat



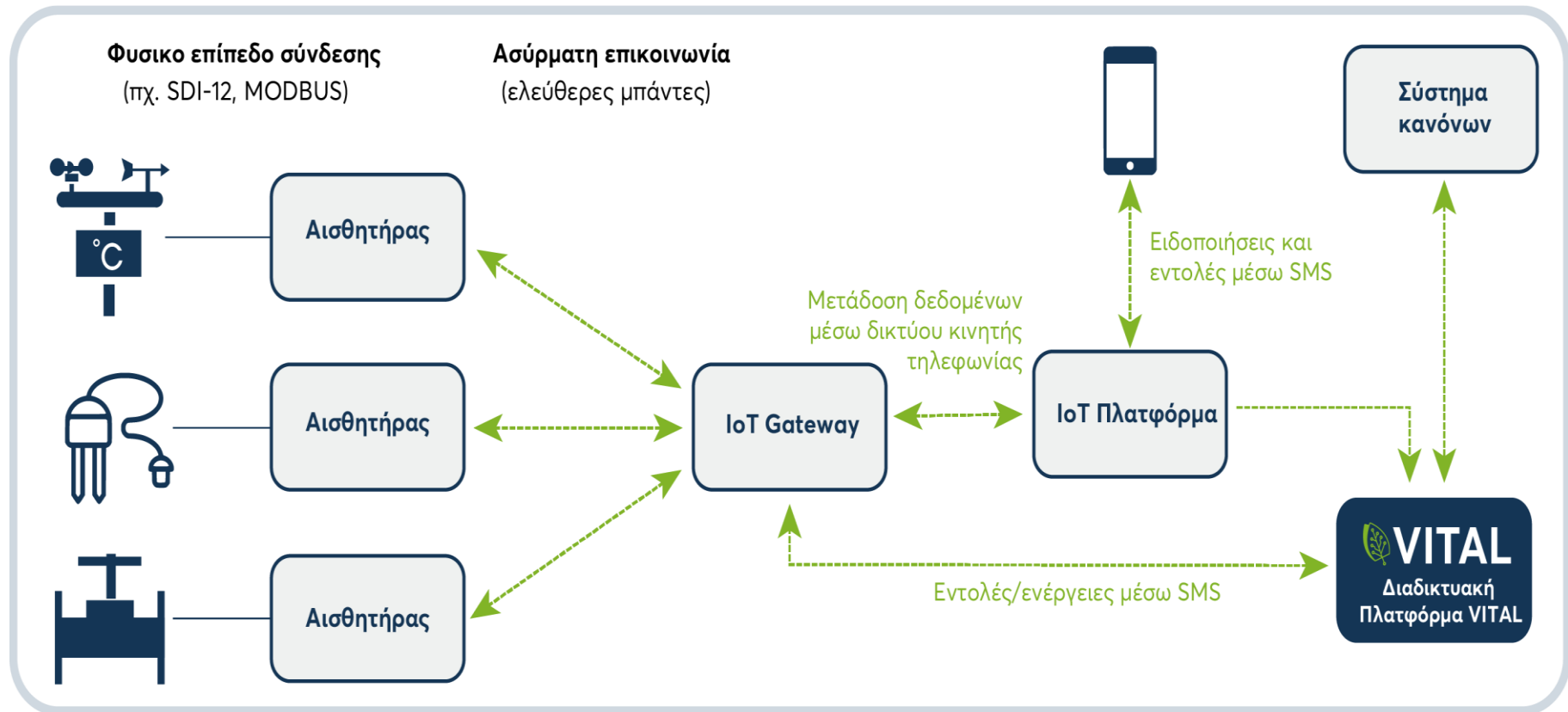
Industrial IoT - Predictive Maintenance (2)



An inclusive overview



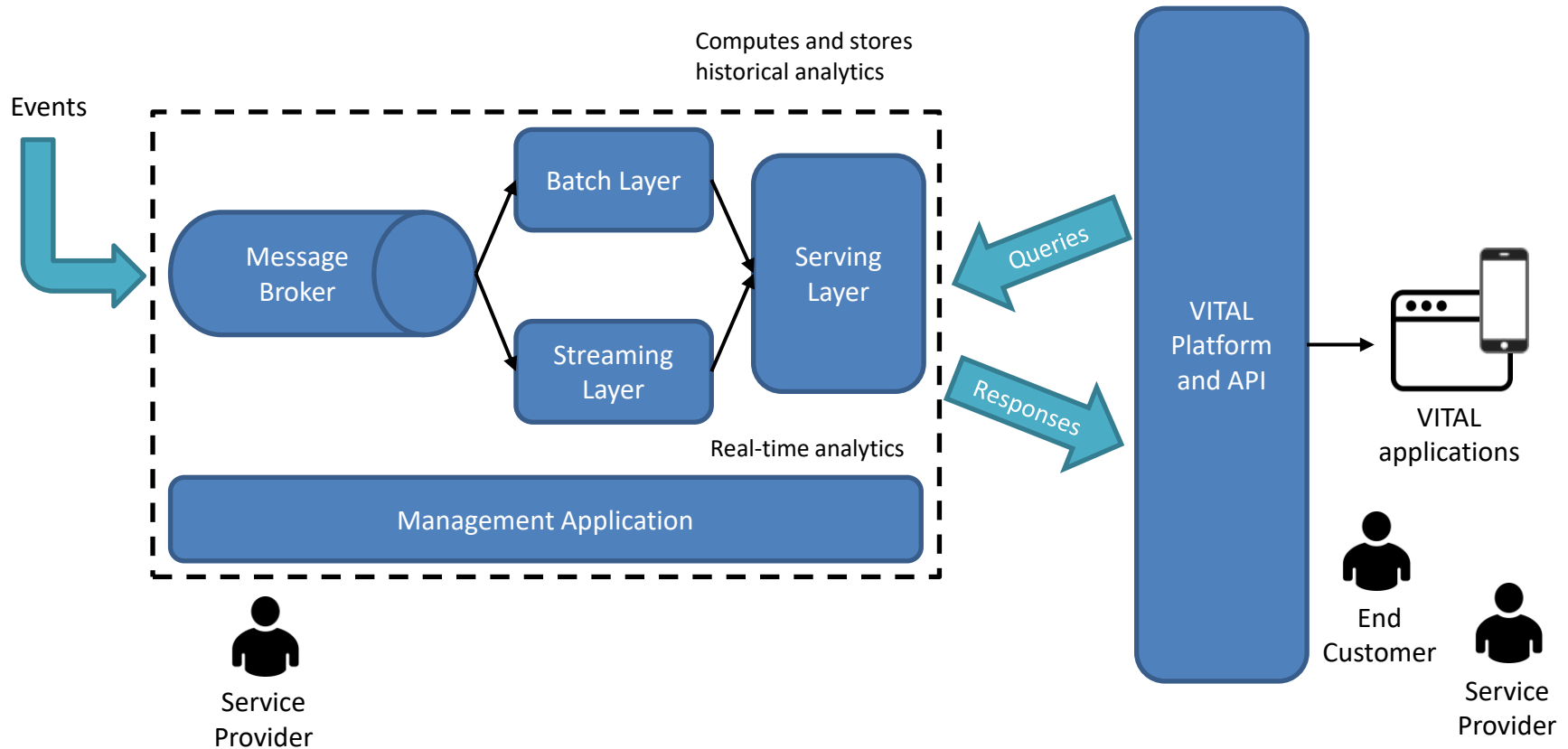
VITAL Project – Overall architecture



National project

Versatile Internet of Things for Agriculture – (VITAL)

VITAL Project – IoT Platform



VITAL Project – Dashboard view (1)





Ilias Ouzounidis
Edit profile | Log out

Dashboard ▾

- Field select
- Overview
- Irrigation
- Smart system
- Data view / export
- Billing

Dashboard

Home > Dashboard > Field 17571 > Overview

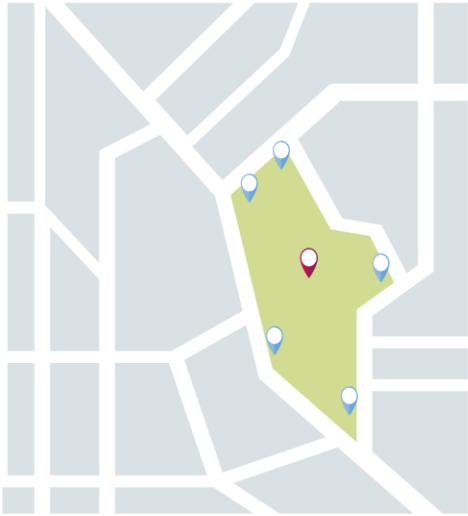
Telemetry

Maps

Field Map

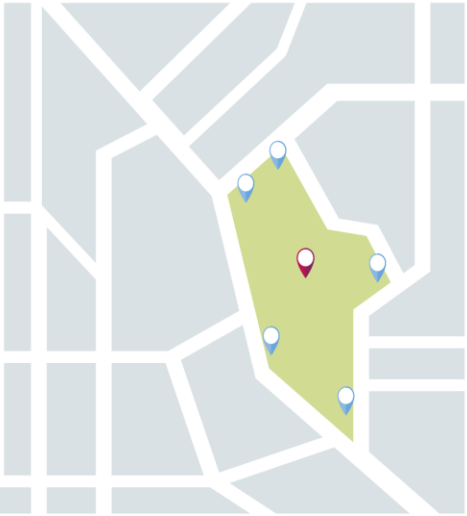
Location of station

Layer Selection ▾
/ / 31



Info

-  Measurement Device
-  Central station





Periklis Chatzimisios, International Hellenic University

page **27**

VITAL Project – Dashboard view (2)



Ilias Ouzounidis
Edit profile | Log out

Dashboard

- Field select
- Overview
- Irrigation
- Smart system
- Data view / export
- Billing

Dashboard

Home > Dashboard > Field 751 > Irrigation

Field information

Crop type: Current type Change

Last reported value of soil moisture
Thu 15/10/2019 10:05:10, 12%

Irrigation Thresholds

Low: 3 High: 15 Change

Control

Auto: Auto Current status: OFF

Irrigation Now

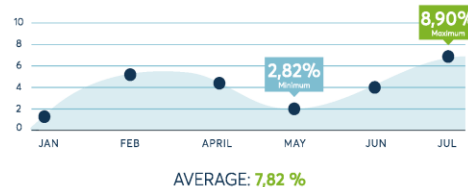
Minutes: 3 Submit

Graphs

From: / / 31

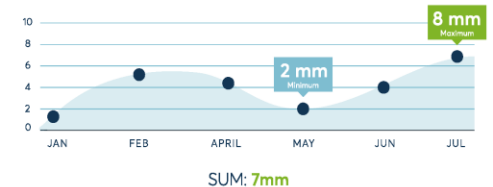
To: / / 31

SOIL MOISTURE



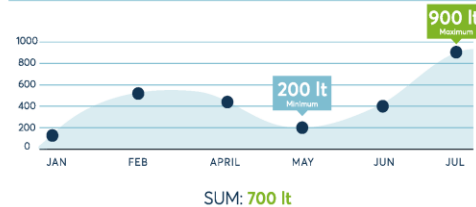
Comments:
Your minimum was May and your Maximum Jul. Something like that

ETc



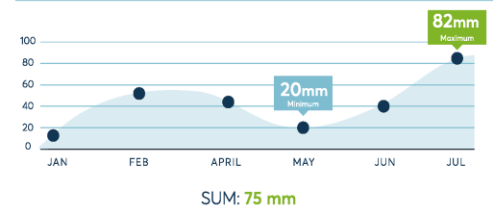
Comments:
Your minimum was May and your Maximum Jul. Something like that

IRRIGATION WATER



Comments:
Your minimum was May and your Maximum Jul. Something like that

RAINFALL



Comments:
Your minimum was May and your Maximum Jul. Something like that



VITAL Project – Dashboard view (3)



Dashboard

Home > Dashboard > Field 751 > Smart system



Ilias Ouzounidis
Edit profile | Log out

Dashboard ▾

- Field select
- Overview
- Irrigation
- Smart system
- Data view / export
- Billing

Smart System

Αναζήτηση 🔍

Overview of rule base of the smart system

+Add Rule

-Delete Rule

Import Rule

Export Rule

IF	Soil Moisture ▾	IS	Greater Than ▾	3 ▴ ▾ +	Activate ON ☒	Select ☐
IF	Soil Moisture ▾	IS	Greater Than ▾	3 ▴ ▾ +	Activate ON ☒	Select ☒
AND	Temperature ▾	IS	Above ▾	5 ▴ ▾		
IF	Soil Moisture ▾	IS	Greater Than ▾	3 ▴ ▾ +	Activate OFF ☐	Select ☐



IIoT scenario: A multi-hop topology

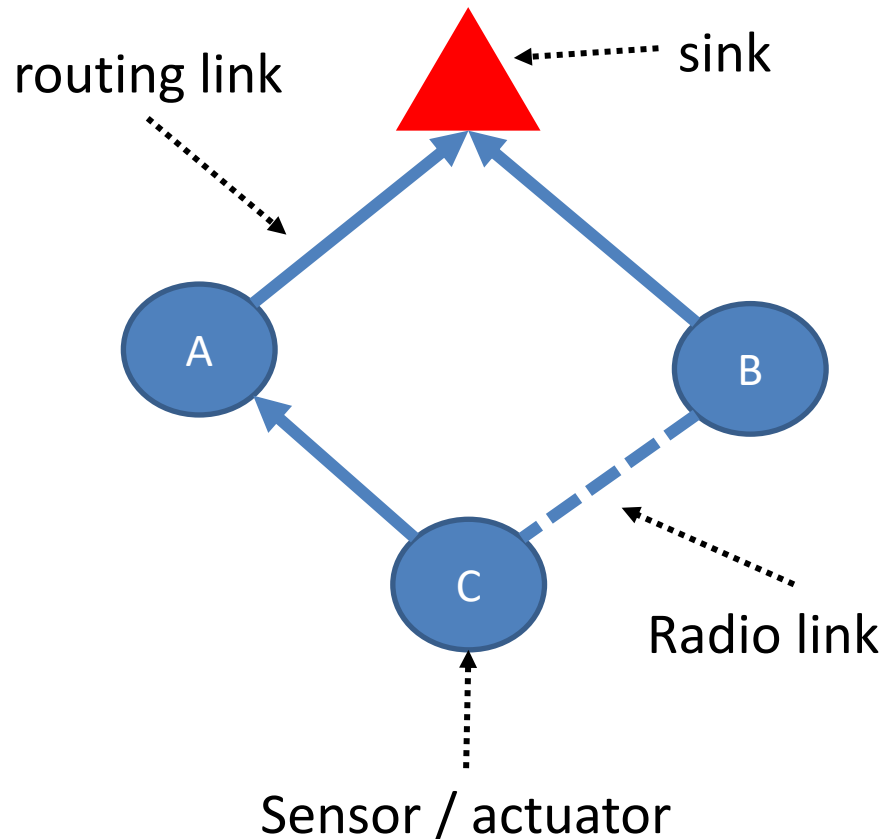
Low-power and Lossy Network (LLN)

Application

- CBR traffic
- converge cast

Environment

- Low power
- Multi-hop
- Lossy links



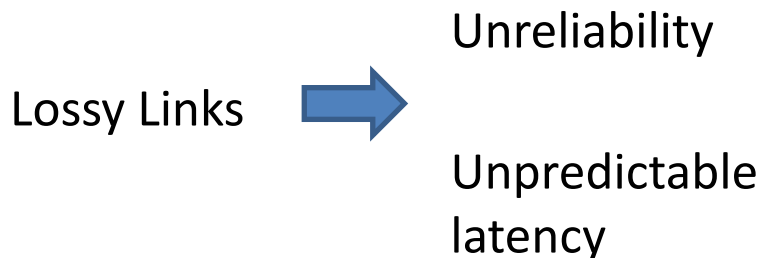
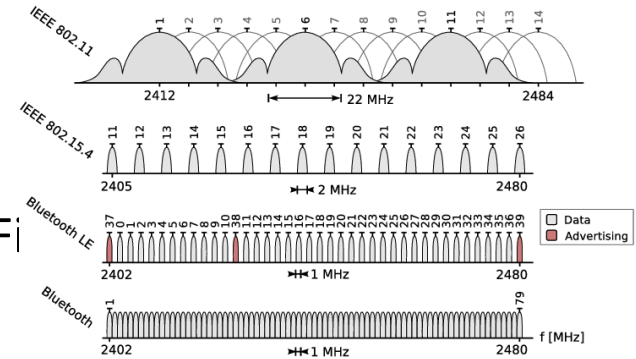
External interference

- Overcrowded ISM band
- Co-located wireless networks
- Co-located high power networks (e.g., Wi-Fi)

Multi-path fading effect

- Reflective structures

Harsh Industrial Environment



MAC Layer

- Coordinating the accesses to shared wireless medium
- Avoiding collisions
- Saving Energy
 - Deafness
 - Idle listening
 - Overhearing

Best effort solutions (CSMA, ALOHA) → Unsuitable for IIoT

Scheduled-based → Deterministic Behavior

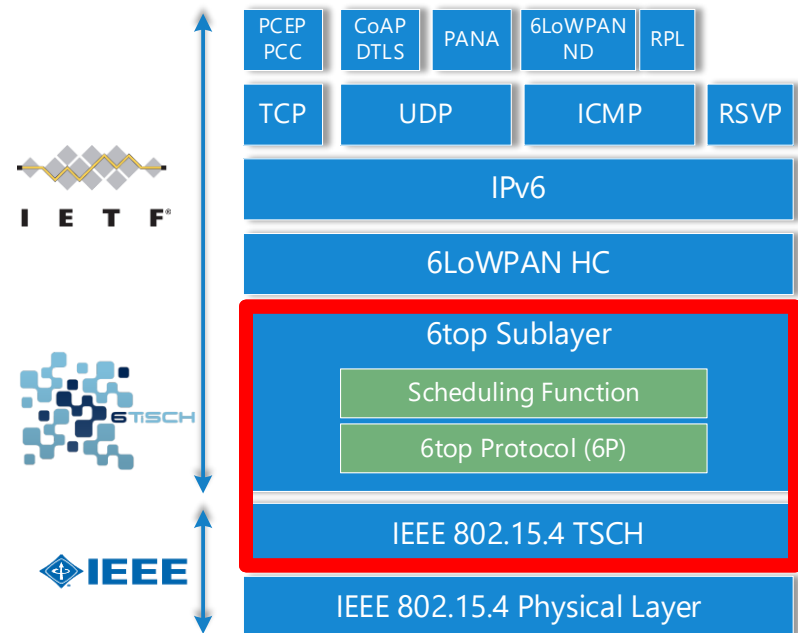
To provide **Reliable Communications for the IIoT**

- Improve/purpose MAC mechanisms to provide
 - High Reliability
 - Low and Deterministic Delay

Research methodology

- Real-world experiments
- Trace based - Simulations

- IEEE
 - Low level protocols
 - Save energy
 - Reliability
- IETF
 - IPv6 Connectivity
- 6TiSCH
 - Glue TSCH, IPv6



Our Research Efforts

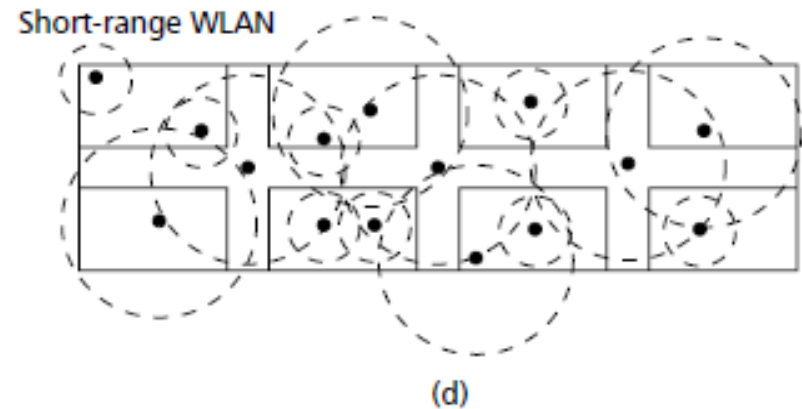
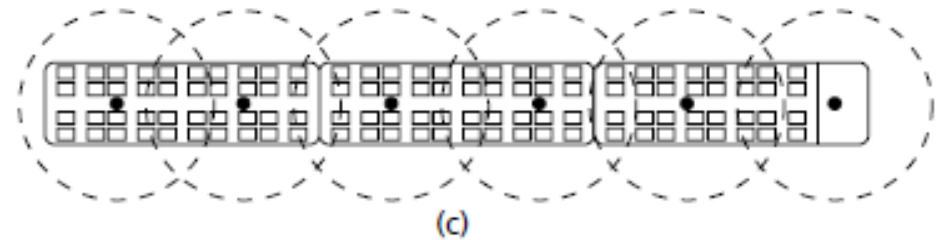
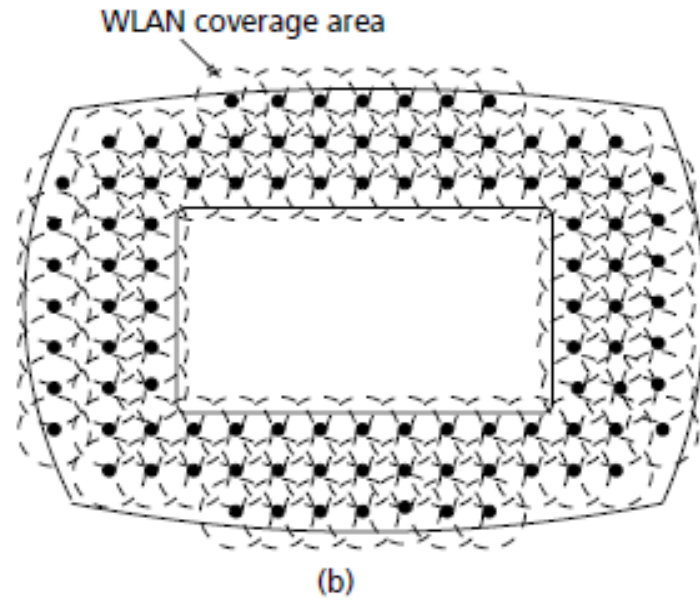
- Enhance Time-Slotted Channel Hopping (TSCH)
- Purpose a Scheduling Function for IIoT

A hot research topic: Very dense networks

- Scenarios could be residential environments with a large number of connected users (apartment buildings, residential complexes), offices and other gathering places such as stadiums and concert halls.

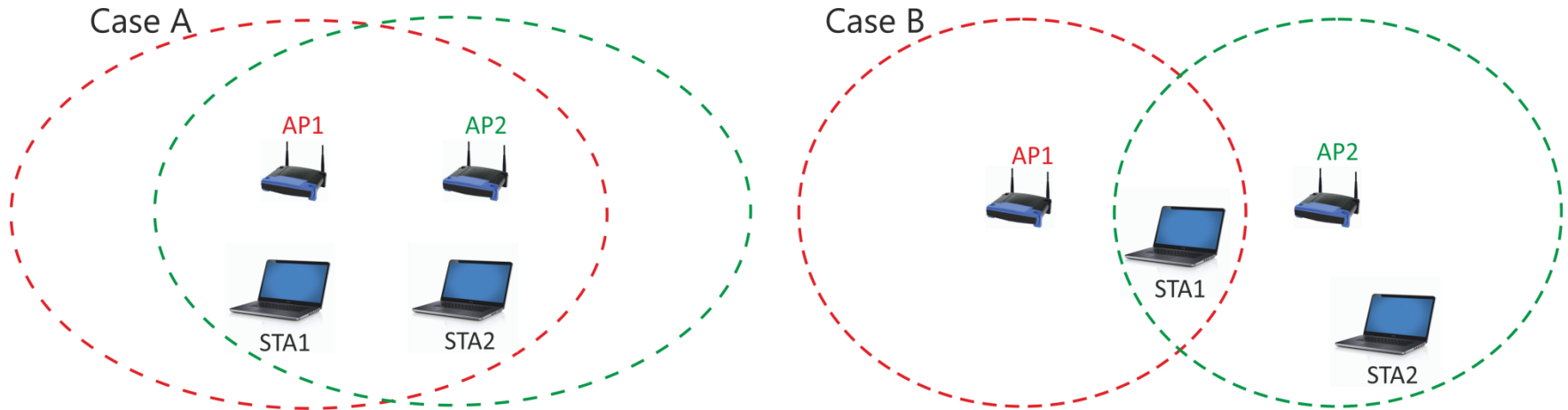


Very dense networks (2)



- Large number of access points and associated STAs deployed in geographical limited region.

Overlapping Basic Service Set (OBSS)



- **Case A:** The Information Element (for the channel selection procedure) is exchanged directly between APs that are in direct range of each other
- **Case B:** The AP can use the neighbor report capability (802.11k amendment) and can request from its associated STAs to scan the medium for neighboring APs and send back a Beacon Report

For addressing the mentioned challenges

- IEEE 802.11 High Efficiency WLANs (HEW) Study Group (SG) to eventually develop an amendment

Main goals of amendment

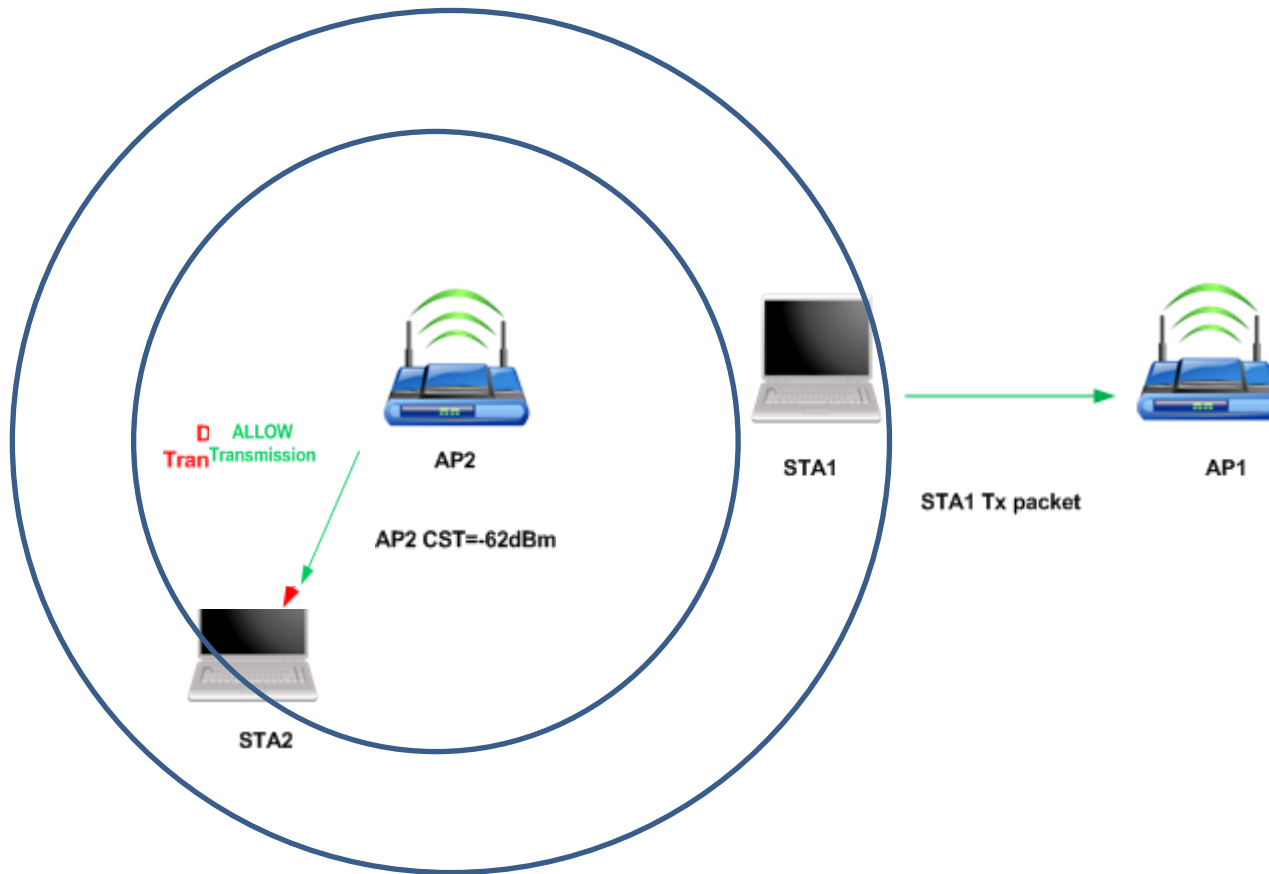
- Improve efficiency in dense networks and efficiency/robustness in outdoor deployments
- Improve power efficiency
- Enabling backward compatibility and coexistence with legacy IEEE 802.11 devices operating in the same band

Efficient parallel utilization of spectrum resources from overlapped WLANs

Expected mechanisms for improving spatial reuse by IEEE 802.11ax

- **Coloring**
 - A technique that can identify the packet Basic Service Set (BSS) origination and type of link (upload or download) in a WLAN
 - We can apply different channel access rules according to coloring information
- **Adaptive carrier sense threshold**
- **Adaptive transmitter power**

Adaptive carrier sense threshold



- CST level for IEEE 802.11 protocols is static so far
- Reduction of the CST level reduces the SR (exposed node) while an increase results in more collisions (hidden node)

Adaptive transmit power

Transmit power (Tx) level for IEEE 802.11 protocols is static so far

Unnecessary high Tx power level from stations or Access Points increases interference to adjacent overlapped WLANs and power consumption

Reducing the Tx power can reduce SNR and data throughput, thus, the proper adjustment is a challenge

Spatial reuse TG is considering the implementation of:

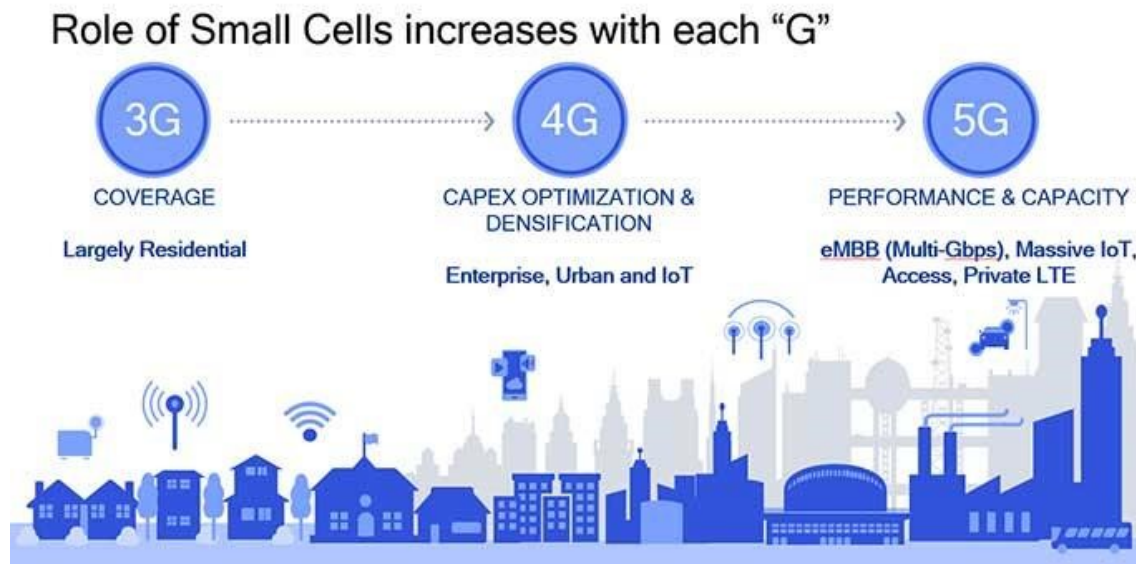
- **Dynamic Sensitivity Control (DSC)**
 - Implements coloring and adaptive carrier sense threshold techniques
 - Applies two different CST levels:
 - Static for Intra_WLAN packets (CST)
 - Dynamic for Inter_WLAN packets (Receiver Strength Signal Indication - RSSI is the reference level for adjusting OBSS threshold)

Project TeamUp5G – Main focus on Small cells

H2020 Marie Curie ITN project “TeamUp5G: A Multidisciplinary Approach to Training and Research on New RAN Techniques for 5G Ultra-Dense Mobile Networks”

SCs are formed by low-power and reduced-size wireless access points that operate in licensed spectrum and are operator-managed.

Due to their low power transmission, SCs can potentially contribute to up to a 60% reduction of power consumption in the global communications infrastructure.



(source: Qualcomm)

The ultra-dense networks with small-cells (SCs) are

- critical to fulfil the user satisfaction
- subject of challenging requirements for the evolution to 5G - must be able to meet the requirements of **very different types of applications**.

Nowadays SC solutions have inherent technical limitations

Possibly the most important challenge is the demand of increasing data rates per km².



UDNs in the new 5G context must be able to meet stringent requirements

- broadband mobile access



- the ecosystem of the Internet of Things (IoT)



This leads to the explosion of SCs that will require new solutions to mitigate interference and optimise the network resources

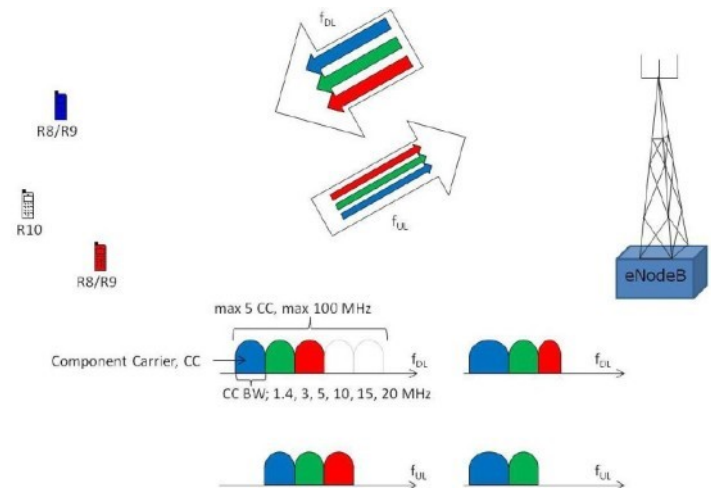
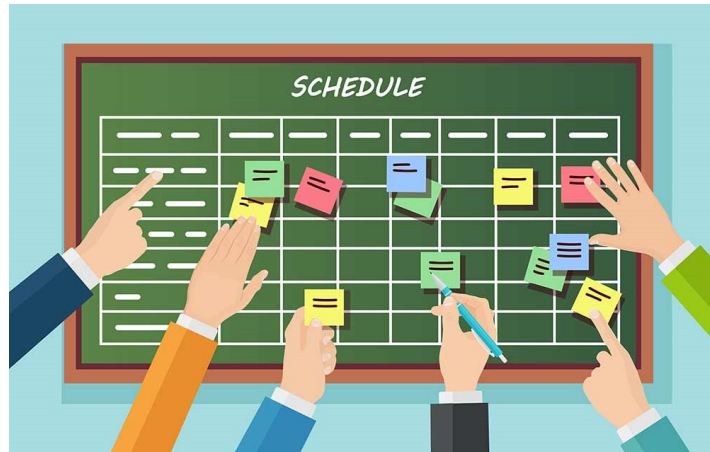
Solutions for research: Dynamic Spectrum Management and Optimisation

Scheduling and resource allocation become essential components of wireless data systems because different users experience different fading conditions

Unlicensed and licensed shared spectrum offer a set of possible advantages, which can potentially lead to greater spectral efficiency than exclusive access

Optimization of the access through the use of shared bands and cognitive radio (CR)

Dynamic spectrum sharing licensed assisted access (LAA) has been adopted by 3GPP, and LAA has adopted CA as a mandatory function



Energy Consumption Reduction

Mobile network energy consumption is very significant

When mixtures of macro and SCs are deployed, the potential of improving becomes even higher

Challenging traffic for emerging 5G applications, multi-connectivity

Further work is needed on mechanisms for energy consumption reduction in the context of “energy-challenged” scenarios such as IoT, particularly considering the increased durations of batteries



Solutions for research: Dynamic Spectrum Management and Optimisation

The project proposes energy-aware MAC protocols with cognitive radio capabilities and optimisations

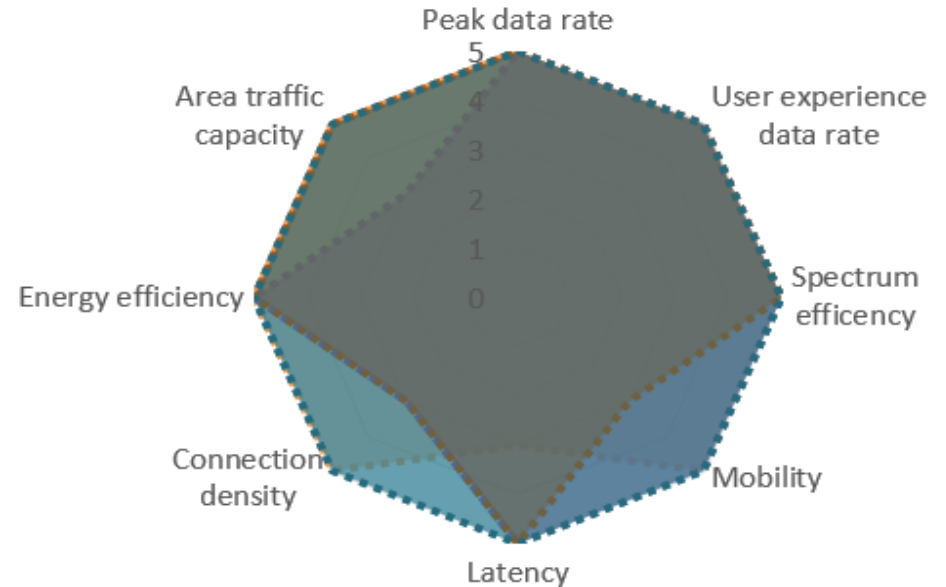
Energy-awareness will be studied for long-range radio technologies

Provision M2M traffic serving as backhaul of the network of sensors along with the own cellular traffic



TeamUp5G Use cases

- Emergency Drone
- Predictive Maintenance
- Multiplayer Game and Live events with VR
- Safe Family Traveling



5G / IoT challenges (summary)

- **Heterogeneity – Interoperability**
- **Scalability – Big Data**
- **Identification and Addressing**
- **Security – Privacy – Trust**
- **Miniaturization of devices**
- **Energy efficiency – Greening of IoT**
- **Standardization**

Thank you!

